STAC67: Regression Analysis

Lecture 12

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Review of Lecture 11

At a new point $\boldsymbol{X}_h$, the mean response is estimated by $\hat{Y}_h = \boldsymbol{X}_h'\hat{\boldsymbol{\beta}}$, with

$$s^2\{\hat{Y}_h\} = MSE\, \boldsymbol{X}_h'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}_h, \qquad s^2\{\text{pred}\} = MSE\left(1+\boldsymbol{X}_h'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}_h\right).$$

Least squares is the orthogonal projection of $\boldsymbol{Y}$ on the column space $C(\boldsymbol{X})$:

$$\hat{\boldsymbol{Y}} = \boldsymbol{H}\boldsymbol{Y}, \qquad \boldsymbol{e} = (\boldsymbol{I}-\boldsymbol{H})\boldsymbol{Y} \perp C(\boldsymbol{X}), \qquad \boldsymbol{X}'\boldsymbol{e} = \boldsymbol{0} \text{ (the normal equations).}$$

$\boldsymbol{H}$ is symmetric and idempotent, with $\boldsymbol{H}\boldsymbol{X} = \boldsymbol{X}$. $\boldsymbol{I}-\boldsymbol{H}$ projects on the $(n-p)$-dimensional space orthogonal to $C(\boldsymbol{X})$, where $\boldsymbol{e}$ lives.

Today: we finish Lecture 11 (the residuals, Pythagoras, degrees of freedom), then tests and intervals, and the overall F test.

Residuals and the analysis of variance

Properties of the residuals

$\boldsymbol{e} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y}$ is again a linear function of $\boldsymbol{Y}$. Since $(\boldsymbol{I} - \boldsymbol{H})\boldsymbol{X} = \mathbf{0}$:

$$E(\boldsymbol{e}) = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{X}\beta = \mathbf{0}, \qquad \boldsymbol{\sigma^2}\{\boldsymbol{e}\} = \sigma^2(\boldsymbol{I} - \boldsymbol{H}),$$

and with normal errors, $\boldsymbol{e} \sim N\big(\mathbf{0},\ \sigma^2(\boldsymbol{I} - \boldsymbol{H})\big)$.

The proof of $\boldsymbol{\sigma^2}\{e\}$ is Assignment 2, Question 3(c).

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Write h_{ii} for the diagonal entries of $\boldsymbol{H}$. On the diagonals of $\boldsymbol{\sigma^2}\{\hat{\boldsymbol{Y}}\} = \sigma^2\boldsymbol{H}$ and of $\boldsymbol{\sigma^2}\{\boldsymbol{e}\}$:

$$Var(\hat{Y}_i) = h_{ii}\,\sigma^2 \ \geq\ 0, \qquad Var(e_i) = (1 - h_{ii})\,\sigma^2 \ \geq\ 0.$$

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$$Var(\hat{Y}_i) = h_{ii}\,\sigma^2 \ \geq\ 0, \qquad Var(e_i) = (1 - h_{ii})\,\sigma^2 \ \geq\ 0.$$

Hence $0 \leq h_{ii} \leq 1$ for every i.

Fitted values and residuals share the variance

Adding the two variance-covariance matrices:

$$\boldsymbol{\sigma^2}\{\hat{\boldsymbol{Y}}\} + \boldsymbol{\sigma^2}\{\boldsymbol{e}\} = \sigma^2\boldsymbol{H} + \sigma^2(\boldsymbol{I} - \boldsymbol{H}) = \sigma^2\boldsymbol{I} = \boldsymbol{\sigma^2}\{\boldsymbol{Y}\}.$$

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For each observation:

$$\underbrace{Var(Y_i)}_{\sigma^2} = \underbrace{Var(\hat{Y}_i)}_{h_{ii}\,\sigma^2} + \underbrace{Var(e_i)}_{(1-h_{ii})\,\sigma^2}.$$

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Since $\hat{Y}_i = \sum_j h_{ij} Y_j$, h_{ii} is the weight of Y_i in its own fitted value.

- With a large h_{ii}, the fit is pulled towards Y_i: $\hat{Y}_i$ varies a lot and e_i little.
- With a small h_{ii}, the opposite.

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- With a small h_{ii}, the opposite.

The residuals do not all have the same variance, even when the errors do.

Exercise: a model with only an intercept

Consider the model

$$Y_i = \beta_0 + \epsilon_i, \qquad i = 1, \ldots, n.$$

1. What is the design matrix $\boldsymbol{X}$?
2. What is the value of h_{ii}, for any i?
3. What is the variance of $\hat{Y}_i$?
4. This is a well-known formula for the variance of an estimator. Which one?
5. Is this result surprising? Explain.

Exercise: solution

1. $\boldsymbol{X} = \mathbf{1}$, the column of n ones, so $\boldsymbol{X}'\boldsymbol{X} = \mathbf{1}'\mathbf{1} = n$ and $(\boldsymbol{X}'\boldsymbol{X})^{-1} = 1/n$.

Exercise: solution

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2. $\boldsymbol{H} = \mathbf{1}\,\frac{1}{n}\,\mathbf{1}' = \frac{1}{n}\boldsymbol{J}$, where $\boldsymbol{J} = \mathbf{1}\mathbf{1}'$ is the $n \times n$ matrix of ones. Every entry of $\boldsymbol{H}$ is $1/n$: $h_{ii} = 1/n$.

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4. It is $Var(\bar{Y}) = \sigma^2/n$: indeed $\hat{\boldsymbol{Y}} = \frac{1}{n}\boldsymbol{J}\boldsymbol{Y} = \bar{Y}\mathbf{1}$, so $\hat{Y}_i = \bar{Y}$ for every i.

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5. Not surprising: least squares minimizes $\sum_i (Y_i - \beta_0)^2$, and setting the derivative $-2\sum_i (Y_i - \beta_0)$ to zero gives $\hat{\beta}_0 = \bar{Y}$. With no predictor, every fitted value is the sample mean.

Pythagoras and the analysis of variance

Since $\mathbf{1}$ is a column of $\boldsymbol{X}$, the vector $\bar{Y}\mathbf{1}$ is in $C(\boldsymbol{X})$, and so is $\hat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}$. Then

$$\boldsymbol{Y} - \bar{Y}\mathbf{1} = (\hat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}) + \boldsymbol{e}, \qquad \text{with} \quad (\hat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}) \perp \boldsymbol{e}.$$

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Pythagoras:

$$\underbrace{\|\boldsymbol{Y} - \bar{Y}\mathbf{1}\|^2}_{SST} = \underbrace{\|\hat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}\|^2}_{SSR} + \underbrace{\|\boldsymbol{e}\|^2}_{SSE}.$$

This is the decomposition of Lecture 6, now for any number of predictors.

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As in Lecture 6, the coefficient of determination

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

is the proportion of the variation of Y explained by the regression, now with any number of predictors.

Degrees of freedom are dimensions

Each sum of squares is the squared length of a vector that lives in a subspace:

Vector	lives in a space of dimension	degrees of freedom
$\boldsymbol{Y} - \bar{Y}\mathbf{1}$	$n-1$ (orthogonal to $\mathbf{1}$)	$n-1$ for SST
$\hat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}$	$p-1$ ($C(\boldsymbol{X})$, orthogonal to $\mathbf{1}$)	$p-1$ for SSR
$\boldsymbol{e}$	$n-p$ (orthogonal to $C(\boldsymbol{X})$)	$n-p$ for SSE

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Why orthogonal to $\mathbf{1}$: with $\boldsymbol{J} = \mathbf{1}\mathbf{1}'$, $\bar{Y}\mathbf{1} = \frac{1}{n}\boldsymbol{J}\boldsymbol{Y}$, so $\boldsymbol{Y} - \bar{Y}\mathbf{1} = (\boldsymbol{I} - \frac{1}{n}\boldsymbol{J})\boldsymbol{Y}$, and

$$\mathbf{1}'\left(\boldsymbol{I} - \frac{1}{n}\boldsymbol{J}\right) = \mathbf{1}' - \frac{1}{n}(\mathbf{1}'\mathbf{1})\,\mathbf{1}' = \mathbf{1}' - \mathbf{1}' = \mathbf{0}'.$$

Likewise for $\hat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}$, since $\mathbf{1}'\hat{\boldsymbol{Y}} = \mathbf{1}'\boldsymbol{Y}$ (because $\mathbf{1}'\boldsymbol{e} = 0$).

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Likewise for $\hat{\boldsymbol{Y}} - \bar{Y}\mathbf{1}$, since $\mathbf{1}'\hat{\boldsymbol{Y}} = \mathbf{1}'\boldsymbol{Y}$ (because $\mathbf{1}'\boldsymbol{e} = 0$).

The degrees of freedom add up as the dimensions do: $n - 1 = (p-1) + (n-p)$.
With one predictor, $p = 2$: as in Lecture 6.

In R: the acceleration data

```r
X <- model.matrix(fit2)
e <- resid(fit2); Yhat <- fitted(fit2)
round(c(t(X) %*% e, sum(Yhat * e)), 8)    # all zero
```

```
## [1] 0 0 0 0
```

```r
Y <- autompg$acc
c(SST = sum((Y - mean(Y))^2),
  SSR_plus_SSE = sum((Yhat - mean(Y))^2) + sum(e^2))
```

```
##          SST SSR_plus_SSE
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Next: tests and intervals for the coefficients. The sums of squares come back for the overall F test.

Inference on one coefficient

Confidence intervals and tests for β_k

With normal errors, $\hat{\boldsymbol{\beta}} \sim N\left(\boldsymbol{\beta}, \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right)$, and $s^2\{\hat{\beta}_k\}$ is the diagonal entry k of $MSE\,(\boldsymbol{X}'\boldsymbol{X})^{-1}$.

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As in simple linear regression, with $n-p$ degrees of freedom instead of $n-2$:

$$\frac{\hat{\beta}_k - \beta_k}{s\{\hat{\beta}_k\}} \sim t(n-p).$$

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- Confidence interval: $\hat{\beta}_k \pm t(1-\alpha/2;\ n-p)\, s\{\hat{\beta}_k\}$.
- Test of $H_0 : \beta_k = 0$ against $H_a : \beta_k \neq 0$: $t^* = \hat{\beta}_k / s\{\hat{\beta}_k\}$, compared with $t(1-\alpha/2;\ n-p)$.

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$H_0 : \beta_k = 0$ asks whether X_k adds anything *once the other predictors are in the model.*

The acceleration data

```
fit2 <- lm(acc ~ wt + hp, data = autompg)
confint(fit2, level = 0.95)
```

```
##                      2.5 %       97.5 %
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## wt           0.001885979  0.002697749
## hp          -0.102212640 -0.084304701
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Among cars of the same weight, 10 more horsepower take between 0.84 and 1.02 seconds off it.

Neither interval contains 0: both predictors matter, each given the other.

Mean responses and new observations

Intervals at a new point X_h

From Lecture 10, $\hat{Y}_h = \boldsymbol{X}_h' \hat{\boldsymbol{\beta}}$ with

$$s^2\{\hat{Y}_h\} = MSE\, \boldsymbol{X}_h'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}_h, \qquad s^2\{\text{pred}\} = MSE + s^2\{\hat{Y}_h\}.$$

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- Confidence interval for the mean response $E(Y_h)$:
  $\hat{Y}_h \pm t(1-\alpha/2;\, n-p)\, s\{\hat{Y}_h\}$.
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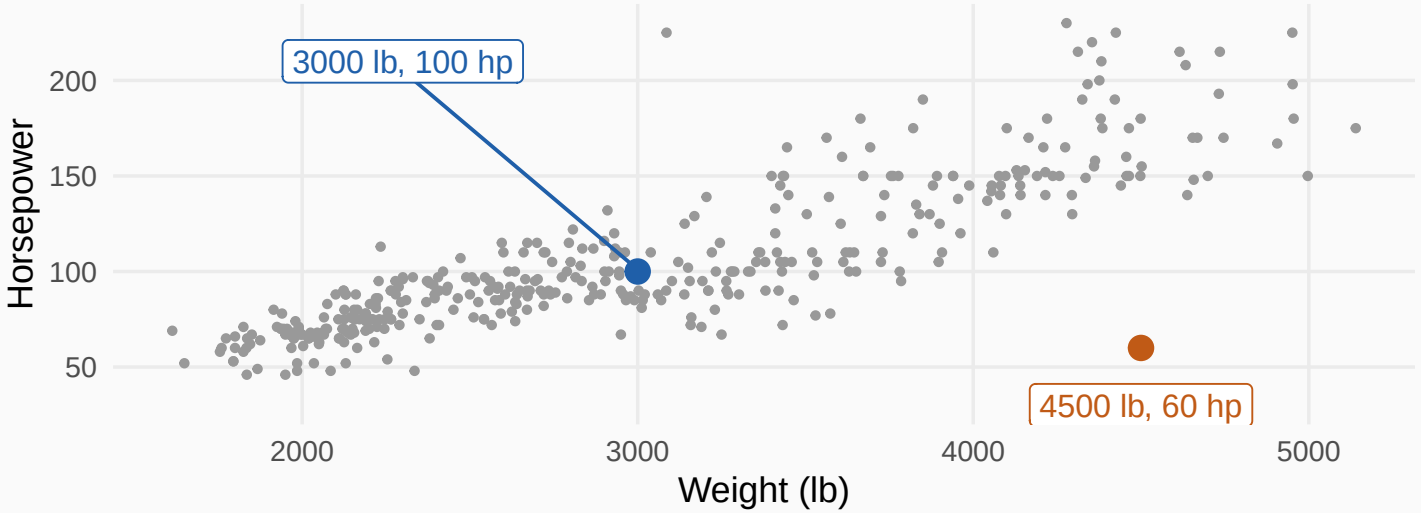
```
new_car <- data.frame(wt = 3000, hp = 100)
predict(fit2, new_car, interval = "confidence")
```

```
##        fit      lwr      upr
## 1 16.01932 15.83937 16.19927
```

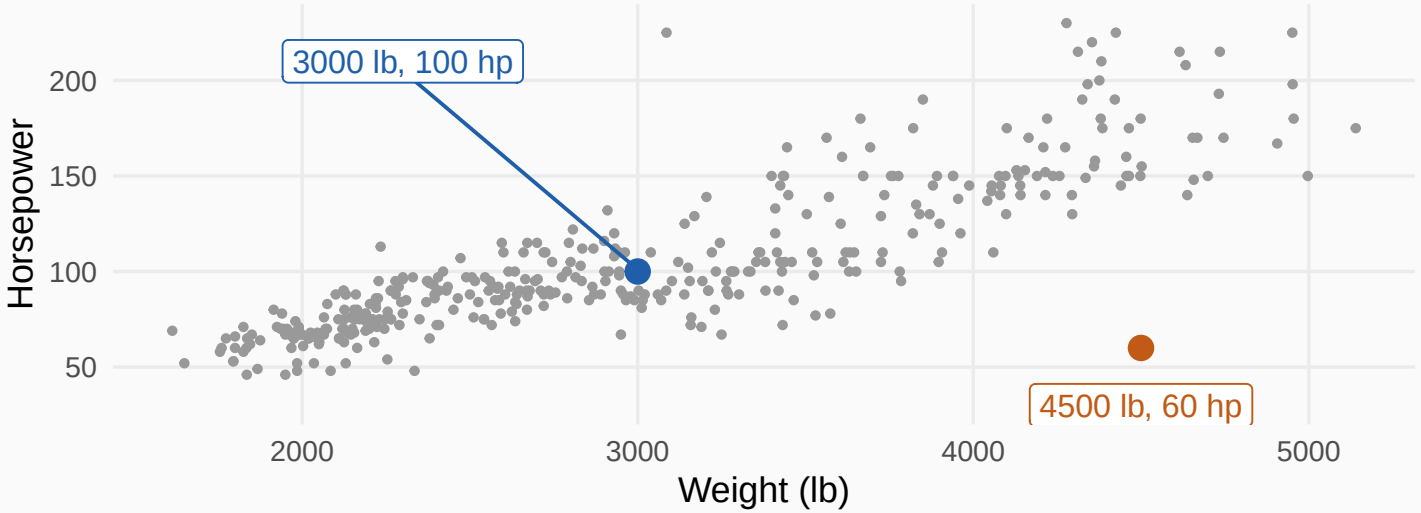
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predict(fit2, new_car, interval = "prediction")
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##        fit      lwr      upr
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Hidden extrapolation

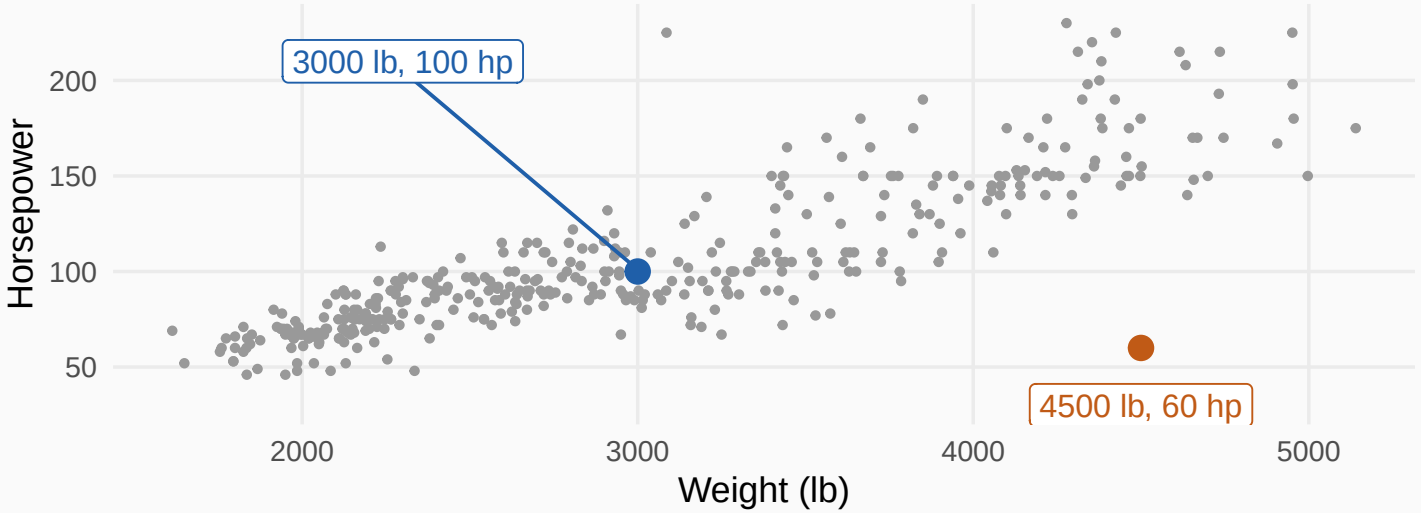


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Its $\boldsymbol{X}_h'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}_h$ is 0.085, against 0.0028 for the car of 3000 lb and 100 hp: its confidence interval is 5.5 times wider, and it rests on the model holding where there are no data.

Analysis of variance

Sums of squares as quadratic forms

Recall $\boldsymbol{J} = \mathbf{1}\mathbf{1}'$, the $n \times n$ matrix of ones, and $\bar{Y}\mathbf{1} = \frac{1}{n}\boldsymbol{J}\boldsymbol{Y}$. Then

$$SST = \boldsymbol{Y}'\left(\boldsymbol{I} - \tfrac{1}{n}\boldsymbol{J}\right)\boldsymbol{Y}, \qquad SSE = \boldsymbol{Y}'(\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y},$$

$$SSR = \boldsymbol{Y}'\left(\boldsymbol{H} - \tfrac{1}{n}\boldsymbol{J}\right)\boldsymbol{Y}.$$

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Each is a quadratic form $\boldsymbol{Y}'\boldsymbol{A}\boldsymbol{Y} = \sum_i \sum_j a_{ij} Y_i Y_j$, with $\boldsymbol{A}$ symmetric.

Sums of squares as quadratic forms

Recall $\boldsymbol{J} = \mathbf{1}\mathbf{1}'$, the $n \times n$ matrix of ones, and $\bar{Y}\mathbf{1} = \frac{1}{n}\boldsymbol{J}\boldsymbol{Y}$. Then

$$SST = \boldsymbol{Y}'\left(\boldsymbol{I} - \tfrac{1}{n}\boldsymbol{J}\right)\boldsymbol{Y}, \qquad SSE = \boldsymbol{Y}'(\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y},$$

$$SSR = \boldsymbol{Y}'\left(\boldsymbol{H} - \tfrac{1}{n}\boldsymbol{J}\right)\boldsymbol{Y}.$$

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Each matrix $\boldsymbol{A}$ here is a projection: symmetric and idempotent.

So $\boldsymbol{Y}'\boldsymbol{A}\boldsymbol{Y} = (\boldsymbol{A}\boldsymbol{Y})'(\boldsymbol{A}\boldsymbol{Y}) = \|\boldsymbol{A}\boldsymbol{Y}\|^2$: the squared length of a projection, as in Lecture 11.

The mean of a quadratic form

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The analogue of $E(Y^2) = Var(Y) + E(Y)^2$.

MSE is unbiased

Apply it to $SSE = \boldsymbol{Y}'(\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y}$, with $\boldsymbol{\Sigma} = \sigma^2\boldsymbol{I}$ and $\boldsymbol{\mu} = \boldsymbol{X}\beta$.

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The trace of a projection is the dimension of the space it projects on: $n - p$, again.

The mean of MSR

For $SSR = \boldsymbol{Y}'(\boldsymbol{H} - \frac{1}{n}\boldsymbol{J})\boldsymbol{Y}$: $tr(\boldsymbol{H} - \frac{1}{n}\boldsymbol{J}) = p - 1$, and since $\boldsymbol{H}\boldsymbol{\mu} = \boldsymbol{\mu}$,

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$$E(MSR) = \sigma^2 + \frac{1}{p-1}\sum_{i=1}^{n}(\mu_i - \bar{\mu})^2 \;\geq\; \sigma^2.$$

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Large values of MSR/MSE are evidence that at least one predictor matters.

The ANOVA table

Source	SS	df	MS	F^*
Regression	SSR	$p-1$	$MSR = SSR/(p-1)$	MSR/MSE
Error	SSE	$n-p$	$MSE = SSE/(n-p)$	
Total	SST	$n-1$		

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With one predictor, $p = 2$, this is the table of Lecture 6.

The overall F test

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0 \qquad \text{against} \qquad H_a : \text{at least one } \beta_k \neq 0.$$

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With normal errors, under H_0, SSR/σ^2 and SSE/σ^2 are independent χ^2 variables with $p-1$ and $n-p$ degrees of freedom (Cochran's theorem, which we take for granted), so

$$F^* = \frac{MSR}{MSE} \sim F(p-1,\ n-p).$$

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⚠ Rejecting says that the predictors, *taken together*, explain some variation. It does not say which ones, nor that the model is good.

The acceleration data

Source	SS	df	MS	F^*
Regression	1794.7	2	897.3	295.7
Error	1174.3	387	3.034	
Total	2969.0	389		

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`anova(fit2)` alone gives a different table, split predictor by predictor: that is next week's extra sums of squares.

R^2 and adjusted R^2

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 0.604 \text{ here}, \qquad R^2 = \left(\text{corr}(\boldsymbol{Y}, \hat{\boldsymbol{Y}})\right)^2.$$

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Adding a predictor can only lower SSE, so R^2 never decreases, even for a useless predictor. Adding a column of pure noise:

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The adjusted R^2 divides each sum of squares by its degrees of freedom:

$$R^2_a = 1 - \frac{SSE/(n-p)}{SST/(n-1)} = 1 - \frac{n-1}{n-p}\frac{SSE}{SST},$$

and can decrease: here from 0.6024 to 0.6014 with the noise.

F and R^2

Dividing the numerator and the denominator of F^* by SST:

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For a fixed R^2, F^* grows with n: with enough data, a weak relation is "significant".

Reading summary(fit2)

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s <- summary(fit2)
round(s$coefficients, 5)
```

```
##             Estimate Std. Error   t value Pr(>|t|)
## (Intercept) 18.46959    0.32651  56.56593        0
## wt           0.00229    0.00021  11.10182        0
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Next week: testing several coefficients at once, extra sums of squares, multicollinearity, qualitative predictors.